

# Daily Geometry Drill #2 — Answers & Investigations

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| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Circle Equations · Tangents · Line–Circle Tests · Chord Lengths.

## Section A Rapid Recognition

1. Write down the centre of the circle  $x^2 + y^2 - 6x - 8y + 24 = 0$ .

**Answer:** "(3,4)"

**Method:** Complete the square:  $(x - 3)^2 + (y - 4)^2 = 25 - 24 = 1$ , so centre (3,4), radius 1.

**Investigate further:** In  $x^2 + y^2 + 2gx + 2fy + c = 0$  the centre is  $(-g, -f)$  and here  $2g = -6$ ,  $2f = -8$ , so  $g = -3$ ,  $f = -4$  — centre (3,4) ✓. **If the coefficients of  $x^2, y^2$  are not 1, divide through first so they are, then read the centre.**

2. Write down the radius of the circle  $x^2 + y^2 - 6x - 8y + 24 = 0$ .

**Answer:** 1

**Method:** From A1,  $(x - 3)^2 + (y - 4)^2 = 1$ , so  $r = \sqrt{1} = 1$ .

**Investigate further:** Formula check:  $r = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 16 - 24} = \sqrt{1} = 1$  ✓. **A "circle" with  $g^2 + f^2 - c < 0$ ? Empty locus — the square sum equals a negative number.**

3. Write down the equation of the circle with centre (2, -1) and radius 5.

**Answer:** " $x^2 + y^2 - 4x + 2y - 20 = 0$ "

**Method:**  $(x - 2)^2 + (y + 1)^2 = 25$  expands to  $x^2 + y^2 - 4x + 2y + 5 = 25$ , i.e.  $x^2 + y^2 - 4x + 2y - 20 = 0$ .

**Investigate further:** Reading back:  $2g = -4 \Rightarrow g = -2$  (centre  $x = 2$ ),  $2f = 2 \Rightarrow f = 1$  (centre  $y = -1$ ),  $r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 1 + 20} = 5$  ✓. **Rewrite for centre (2, -1) radius 2: what becomes of the constant?  $x^2 + y^2 - 4x + 2y + 1 = 0$ .**

4. Write down the radius of the circle  $x^2 + y^2 - 4x + 6y - 12 = 0$ .

**Answer:** 5

**Method:** Complete the square:  $(x - 2)^2 + (y + 3)^2 = 4 + 9 + 12 = 25$ , so  $r = 5$ .

**Investigate further:** Formula:  $g = -2, f = 3, r = \sqrt{4 + 9 + 12} = 5$  ✓. **All these squares are perfect:  $5^2, 4^2, 3^2$ . What radius would an *irrational* answer look like, and would that stop a clean distance test? Only the appearance changes; the distance test still works.**

5. The point  $(x, 4)$  lies on the circle  $x^2 + y^2 = 25$  with  $x > 0$ . Find  $x$ .

**Answer:** 3

**Method:**  $x^2 + 4^2 = 25 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$ ; take +3.

**Investigate further:** Both  $(3, 4)$  and  $(-3, 4)$  lie on the circle — the  $x > 0$  condition kills one branch.

**List all lattice points**  $(x, y)$  **on**  $x^2 + y^2 = 25$ .  $(0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3), (\pm 5, 0)$ .

6. The radius from the origin to the point  $(3, 4)$  on  $x^2 + y^2 = 25$  has what gradient?

**Answer:**  $\frac{4}{3}$

**Method:** Gradient =  $\frac{4-0}{3-0} = \frac{4}{3}$ .

**Investigate further:** The radius is the line joining centre to the point. **Gradient of the perpendicular radius to**  $(3, -4)$ ?  $-\frac{4}{3}$ .

7. The tangent to  $x^2 + y^2 = 25$  at the point  $(3, 4)$  has what gradient?

**Answer:**  $-\frac{3}{4}$

**Method:** Tangent  $\perp$  radius, and radius gradient is  $\frac{4}{3}$ , so tangent gradient =  $-\frac{1}{4/3} = -\frac{3}{4}$ .

**Investigate further:** The tangent line is  $3x + 4y = 25$  (via  $xx_1 + yy_1 = r^2$ ). **Does it pass through**  $(3, 4)$ ?  $9 + 16 = 25$  ✓. **What is its x-intercept?**  $x = 25/3$ .

8. Write down the equation of the circle with centre  $(-1, 2)$  and radius 3.

**Answer:** " $x^2 + y^2 + 2x - 4y - 4 = 0$ "

**Method:**  $(x + 1)^2 + (y - 2)^2 = 9$  expands to  $x^2 + y^2 + 2x - 4y + 5 = 9$ , i.e.  $x^2 + y^2 + 2x - 4y - 4 = 0$ .

**Investigate further:** Check centre:  $g = 1 \Rightarrow$  centre  $x = -1$ ,  $f = -2 \Rightarrow$  centre  $y = 2$  ✓;  $r = \sqrt{1 + 4 + 4} = 3$  ✓. **Where does this circle cross the axes?** Put  $y = 0$ :  $x^2 + 2x - 4 = 0$ , a surd — compare with the clean integer axes of Day 1 right-triangles.

9. The centre of  $x^2 + y^2 - 6x - 8y + 24 = 0$  is  $(3, 4)$ . Find the perpendicular distance of this centre from the line  $4x + 3y - 9 = 0$ .

**Answer:** 3

**Method:** Distance =  $\frac{|4(3) + 3(4) - 9|}{\sqrt{4^2 + 3^2}} = \frac{|12 + 12 - 9|}{5} = \frac{15}{5} = 3$ .

**Investigate further:** This is day 1's perpendicular-distance formula applied to a circle's centre. **Since**  $d = 3 > r = 1$ , **the line misses the circle. How many common tangents could such an external line family have?** Preview of Day 3's tangent-counting.

10. A chord of the circle  $x^2 + y^2 = 25$  is at perpendicular distance 3 from the origin. Find the length of the chord.

**Answer:** 8

**Method:** Half-chord =  $\sqrt{r^2 - d^2} = \sqrt{25 - 9} = \sqrt{16} = 4$ ; full length = 8.

**Investigate further:** The perpendicular from the centre to a chord bisects it — the half-chord

formula. **If the distance were 4, the length is 6; if 5, it degenerates to a tangent. Check both.** The chord shrinks to a point exactly when  $d = r$ .

## Section B Manipulation Drills

1. Which of the following is the equation of the circle with centre  $(3, -2)$  and radius 4?

A)  $x^2 + y^2 - 6x + 4y - 3 = 0$

B)  $x^2 + y^2 - 6x + 4y + 9 = 0$

C)  $x^2 + y^2 + 6x - 4y - 3 = 0$

D)  $x^2 + y^2 - 6x - 4y - 3 = 0$

**Answer:** A)  $x^2 + y^2 - 6x + 4y - 3 = 0$

**Method:**  $(x - 3)^2 + (y + 2)^2 = 16$  gives  $x^2 + y^2 - 6x + 4y + 13 = 16$ , i.e.  $x^2 + y^2 - 6x + 4y - 3 = 0$ .

**Investigate further:** Only A has both the  $x$ -coefficient  $-6$  (centre  $x = 3$ ) and the  $y$ -coefficient  $+4$  (centre  $y = -2$ ). **B has the right coefficients but the wrong constant  $+9$ ; what radius would that give?  $r = \sqrt{9 + 4} = 5$ .**

2. The circle  $x^2 + y^2 + 2x - 4y - 11 = 0$  has what radius?

A) 2

B) 3

C) 4

D)  $\sqrt{5}$

**Answer:** C) 4

**Method:**  $(x + 1)^2 + (y - 2)^2 = 1 + 4 + 11 = 16$ , so  $r = 4$ .

**Investigate further:** Complete the square exactly:  $x^2 + 2x = (x + 1)^2 - 1$  and  $y^2 - 4y = (y - 2)^2 - 4$ , so  $(x + 1)^2 + (y - 2)^2 = 11 + 1 + 4 = 16$  and  $r = 4$  ✓. With  $g = 1, f = -2, c = -11$  the formula reads  $r = \sqrt{g^2 + f^2 - c} = \sqrt{1 + 4 + 11}$ . **Option D ( $\sqrt{5}$ ) traps students who drop the constant 11 (leaving  $r^2 = 1 + 4 = 5$ ) or flip the sign of  $c$ ; keep the sign fixed through the whole square.**

3. The point  $(1, 2)$  lies on the circle  $x^2 + y^2 = 5$ . The equation of the tangent to the circle at  $(1, 2)$  is?

A)  $y = -\frac{1}{2}x + \frac{5}{2}$

B)  $y = 2x$

C)  $y = -2x + 5$

D)  $y = \frac{1}{2}x$

**Answer:** A)  $y = -\frac{1}{2}x + \frac{5}{2}$

**Method:** Radius gradient  $= \frac{2}{1} = 2$ ; tangent gradient  $= -\frac{1}{2}$ ; through  $(1, 2)$ :  $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$ .

**Investigate further:** Option C uses the wrong sign ( $-2x$  is the negative-reciprocal error recalling Day 1), B is the tangent being perpendicular to the perpendicular. **Split-variable test: tangent to  $x^2 + y^2 = r^2$  at  $(x_1, y_1)$  is  $xx_1 + yy_1 = r^2$ : here  $x + 2y = 5 \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$  ✓.**

4. A circle of form  $x^2 + y^2 + 2gx + 2fy + c = 0$  passes through the origin. Deduce  $c$ .

- A)  $c = 1$   
 B)  $c = 0$   
 C)  $c = g + f$   
 D)  $c = -1$

**Answer:** B)  $c = 0$

**Method:** At the origin  $(0, 0) \in$  circle means  $0 + 0 + 0 + 0 + c = 0$  — the constant term is the value of the LHS at the origin, so  $c = 0$ .

**Investigate further:** Circle  $x^2 + y^2 = 25$  has  $c = -25$  and indeed does *not* pass through the origin. The constant records the origin's membership:  $c = 0 \Leftrightarrow$  origin on the circle.

5. The perpendicular distance from the centre of  $x^2 + y^2 = 25$  to the line  $y = 2x + 3$  is?

- A)  $\frac{3}{\sqrt{5}}$   
 B)  $\frac{3}{5}$   
 C) 3  
 D) 2

**Answer:** A)  $\frac{3}{\sqrt{5}}$

**Method:** Line  $2x - y + 3 = 0$ ; distance from  $(0, 0) = \frac{|3|}{\sqrt{2^2 + (-1)^2}} = \frac{3}{\sqrt{5}}$ .

**Investigate further:** Compare with  $r = 5$ :  $\frac{3}{\sqrt{5}} < 5$ , so the line cuts the circle. This is exactly the Day-1 formula put to the new line-circle test.

6. Which of the following is a tangent to the circle  $x^2 + y^2 = 4$ ?

- A)  $y = x + 2\sqrt{2}$   
 B)  $y = x + 4$   
 C)  $y = 2x + 2$   
 D)  $y = 2x + 4$

**Answer:** A)  $y = x + 2\sqrt{2}$

**Method:** Line  $x - y + 2\sqrt{2} = 0$  has distance from origin  $\frac{2\sqrt{2}}{\sqrt{2}} = 2 = r$  — tangent. The others give  $\frac{4}{\sqrt{2}} = 2\sqrt{2}$ ,  $\frac{2}{\sqrt{5}}$ ,  $\frac{4}{\sqrt{5}}$ , none equal to 2.

**Investigate further:** Only option A places the origin exactly  $r$  away. **Check B:**  $y = x + 4$  has distance  $2\sqrt{2} > 2$ , so it misses the circle. Shifting a tangent line a little breaks tangency —

| distance must *equal*  $r$ .

7. A chord of the circle  $x^2 + y^2 = 13$  has length  $4\sqrt{3}$ . What is the perpendicular distance of the chord from the centre?

- A) 3  
B) 2  
C)  $\sqrt{5}$   
D) 1

**Answer:** D) 1

**Method:** Half-chord =  $2\sqrt{3}$ ;  $(2\sqrt{3})^2 = 13 - d^2 \Rightarrow 12 = 13 - d^2 \Rightarrow d^2 = 1 \Rightarrow d = 1$ .

**Investigate further: Fallback: remember**  $(\text{half-chord})^2 + (\perp \text{ dist})^2 = r^2$ . Statement C ( $\sqrt{5}$ ) is the trap of  $13 - 8$  instead of  $13 - 12$ .

8. The line  $y = mx + 4$  is a tangent to the circle  $x^2 + y^2 = 4$ . Find the two possible values of  $m$ .

- A)  $\pm 1$   
B)  $\pm\sqrt{3}$   
C)  $\pm 2$   
D)  $\pm\sqrt{2}$

**Answer:** B)  $\pm\sqrt{3}$

**Method:** Line  $mx - y + 4 = 0$ ; tangency requires  $\frac{|4|}{\sqrt{m^2+1}} = 2 \Rightarrow \sqrt{m^2+1} = 2 \Rightarrow m^2 = 3 \Rightarrow m = \pm\sqrt{3}$ .

**Investigate further:** Both tangent lines pass through  $(0, 4)$ , symmetric about the  $y$ -axis — the two tangents from an external point. **How long is each tangent segment from  $(0, 4)$  to its touch point?**  $\sqrt{16 - 4} = \sqrt{12} = 2\sqrt{3}$  — a preview of Power of a Point (Day 5).

9. The circle with centre  $(6, a)$  and radius 6 is tangent to the  $y$ -axis and passes through the point  $(0, 8)$ . Find  $a$ .

- A) 1  
B) 3  
C) 5  
D) 8

**Answer:** D)  $a = 8$

**Method:** Tangent to the  $y$ -axis ( $x = 0$ ) since centre's  $x$ -distance  $6 = r$ . Through  $(0, 8)$ :  $(0-6)^2 + (8-a)^2 = 36 \Rightarrow 36 + (8-a)^2 = 36 \Rightarrow a = 8$ .

**Investigate further:** Setting  $r = 6$  fixes the  $x$ -distance; passing through  $(0, 8)$  forces the centre level with that point. **Option C (5) assumes the centre is  $(6, a)$  with  $(8-a) = 1$ ?** Check  $(8-5)^2 = 9 \neq 0$ .

10. Write down the length of the chord cut from the line  $y = x + 1$  by the circle  $x^2 + y^2 = 25$ .

**Answer:**  $7\sqrt{2}$

**Method:** Line  $x - y + 1 = 0$ ; distance from the origin  $= \frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$ . Half-chord  $= \sqrt{25 - \frac{1}{2}} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$ ; full chord  $= \frac{14}{\sqrt{2}} = 7\sqrt{2}$ .

**Investigate further:** Note the line does not pass through the centre, so this is a genuine off-centre chord. If it were  $y = x$  the chord would be a full diameter 10.

## Section C Substitution & Structure

( $\leq 90$  s each)

*Harder: hidden tangency, chord bisection, and the distance-vs-discriminant decision. No brute substitution.*

1. A tangent to the circle  $x^2 + y^2 = 144$  passes through  $(20, 0)$  and meets the positive  $y$ -axis at  $T$ . What is the  $y$ -coordinate of  $T$ ? (after TMUA 2017 Paper 1 Q6)

- A) 12  
B) 15  
C)  $\frac{49}{3}$   
D) 20

**Answer:** B) 15

**Method:** Radius to touch point is  $\perp$  tangent. Let  $T = (0, t)$ , line through  $(20, 0)$  and  $(0, t)$  has equation  $x/20 + y/t = 1$ , i.e.  $tx + 20y - 20t = 0$ . Distance from origin  $= r = 12$ :  $|-20t|/\sqrt{t^2 + 400} = 12 \Rightarrow 400t^2 = 144(t^2 + 400) \Rightarrow 256t^2 = 57600 \Rightarrow t^2 = 225 \Rightarrow t = 15$  (positive). Check  $t = 15$  gives line  $3x + 4y = 60$ , distance  $60/5 = 12 \checkmark$ .

**Investigate further:** The  $x$ -intercept 20 and  $r = 12$  give a 5-12-13 scaled triple  $(12, 16, 20)$ . **If the external point were  $(13, 0)$  instead, the  $y$ -int would be  $12 \cdot 13/5$ ?** No, the triple changes.

2. The point  $P(8, 6)$  is outside the circle  $x^2 + y^2 = 36$ . Tangents from  $P$  meet the circle at  $A$  and  $B$ . What is the length  $AB$ ? (adapted from TMUA 2018 Paper 1 Q7)

- A)  $\frac{12\sqrt{13}}{13}$   
B)  $\frac{24\sqrt{13}}{13}$   
C)  $\frac{48}{5}$   
D)  $\frac{72}{5}$

**Answer:** C)  $\frac{48}{5}$

**Method:**  $OP = \sqrt{64 + 36} = 10$ ,  $r = 6$ , right triangle  $OAP$  has  $AP = \sqrt{100 - 36} = 8$ , distance from  $O$  to chord  $AB$  is  $d = r^2/OP = 36/10 = 18/5$ . Half-chord  $= \sqrt{r^2 - d^2} = \sqrt{36 - 324/25} = 24/5$ , so  $AB = 48/5$ .

**Investigate further:** The chord  $AB$  is the radical axis of the circle and the point-circle. **If  $P$  were at distance  $2r$  from  $O$ , the chord length would be  $r\sqrt{3}$  — check. Here  $OP = 10 = 5r/3$ , giving  $48/5$ .**

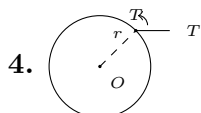
3. The circles  $(x + 4)^2 + (y + 1)^2 = 64$  and  $(x - 8)^2 + (y - 4)^2 = r^2$  have exactly one point in common. If  $r < 13$ , what is  $r$ ? (after TMUA 2019 Paper 1 Q6)

- A) 5  
B) 8  
C) 13  
D) 21

**Answer:** A) 5

**Method:** Centre distance  $d = \sqrt{12^2 + 5^2} = 13$ . One point  $\Rightarrow$  internal tangency  $r = |d - 8| = 5$  (since  $r < 13$ ). External would give 21.

**Investigate further:** Two possibilities:  $|r - 8| = 13$  gives  $r = 5$  or 21; the condition  $r < 13$  picks 5.  
**If  $r > 13$ , the other is 21.**



A circle has centre  $(9, 11)$  and radius 5. A regular hexagon  $ABCDEF$  is inscribed with all vertices on the circle. What is the area of the hexagon? (adapted from TMUA 2017 Paper 1 Q9)

- A)  $18\sqrt{3}$   
B)  $36\sqrt{3}$   
C)  $\frac{75\sqrt{3}}{2}$   
D) 75

**Answer:** C)  $\frac{75\sqrt{3}}{2}$

**Method:** Regular hexagon = 6 equilateral triangles side  $r = 5$ : area =  $6 \cdot \frac{\sqrt{3}}{4}r^2 = \frac{3\sqrt{3}}{2}r^2 = \frac{75\sqrt{3}}{2}$ .

**Investigate further:** Centre  $(9, 11)$  is irrelevant — translation does not change area. **If it were a square, area would be  $2r^2 = 50$ .**

5. The line  $y = 2x + 5$  is a tangent to the circle  $x^2 + y^2 = r^2$ . Which  $r^2$  makes the distance from the centre to the line equal to  $r$ ?

- A) 1  
B) 5  
C) 20  
D) 25

**Answer:** B) 5

**Method:** Distance =  $|5|/\sqrt{5} = \sqrt{5} = r$ , so  $r^2 = 5$ .

**Investigate further:** Tangent to centred circle  $r = |c|/\sqrt{a^2 + b^2}$ . **If line were  $y = 2x + 10$ ,  $r^2$  would be 20.**

6. The circle  $x^2 + y^2 - 4x + 2y - 11 = 0$  and the line  $x - y + 1 = 0$ : determine the number of intersection points without solving the system. (after MAT 2019 Q6)

- A) 0 (line misses)  
 B) 1 (tangent)  
 C) 2 (secant)  
 D) Infinitely many (line is diameter)

**Answer:** C) 2 (secant)

**Method:** Centre  $(2, -1)$ ,  $r = 4$ , distance to  $x - y + 1 = 0$  is  $|2 + 1 + 1|/\sqrt{2} = 4/\sqrt{2} = 2\sqrt{2} < r$ , so 2 points.

**Investigate further:** Compare  $d$  vs  $r$ , never substitute. **Diameter would need  $d = 0$ .**

7.  The circle  $x^2 + y^2 - 8x - 4y - 21 = 0$  ( $r = \sqrt{41}$ ) is cut by the  $y$ -axis ( $x = 0$ ). What is the length of the chord intercepted on the  $y$ -axis?

- A)  $2\sqrt{21}$   
 B)  $2\sqrt{41}$   
 C) 8  
 D) 10

**Answer:** D) 10

**Method:** Centre  $(4, 2)$ , distance to  $x = 0$  is 4, half-chord  $\sqrt{41 - 16} = 5$ , length 10. Also  $x = 0 \Rightarrow y^2 - 4y - 21 = 0 \Rightarrow y = 7, -3$ , distance 10.

**Investigate further:** Two methods: distance vs. direct substitution give 10. **If the line were  $x = 4$  (through centre), chord would be diameter  $2\sqrt{41}$ .**

8. The line  $y = kx + 5$  is tangent to the circle  $x^2 + y^2 - 6x - 8y + 24 = 0$  (centre  $(3, 4)$ ,  $r = 1$ ). How many distinct  $k$  satisfy this? (after TMUA 2020 Paper 1 Q7)

- A) 0  
 B) 1  
 C) 2  
 D) Infinitely many

**Answer:** C) 2

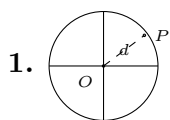
**Method:** Point  $(0, 5)$  is distance  $\sqrt{10} > 1$  from centre, so external: two tangents. Solve  $|3k - 4 + 5|/\sqrt{k^2 + 1} = 1$  gives 2 slopes.

**Investigate further:** From an external point exactly two tangents; from a point on the circle one; from inside zero. **Here  $(0, 5)$  is  $\sqrt{10}$  away, so two.**

## Section D Challenge

(TMUA / SMC / BMO1 difficulty)

*Hardest 5: distance invariants, hidden right angles, and a construction you must spot.*



1. The circles  $C_1 : (x+2)^2 + (y-3)^2 = 18$  and  $C_2 : (x-7)^2 + (y+6)^2 = 2$  have centres  $O_1, O_2$ . What is the *shortest* distance between a point on  $C_1$  and a point on  $C_2$ ? (*adapted from TMUA 2018 Paper 1 Q7 / 2022 Paper 1 Q3*)

- A)  $5\sqrt{2} - 4$   
 B)  $5\sqrt{2} - 5$   
 C)  $5\sqrt{2}$   
 D)  $5\sqrt{2} + 5$

**Answer:** C)  $5\sqrt{2}$

**Method:**  $O_1 = (-2, 3)$ ,  $O_2 = (7, -6)$ ,  $O_1O_2 = \sqrt{9^2 + 9^2} = 9\sqrt{2}$ ,  $r_1 = 3\sqrt{2}$ ,  $r_2 = \sqrt{2}$ , sum  $4\sqrt{2}$ , shortest  $= 9\sqrt{2} - 4\sqrt{2} = 5\sqrt{2}$ . Externally separated ( $d > r_1 + r_2$ ).

**Investigate further:** Shortest = centre distance minus sum of radii (external). If one circle contained the other, it would be  $|r_1 - r_2| - d$ . For these numbers,  $5\sqrt{2} \approx 7.07$ .

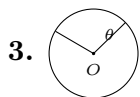
2. The segment joining  $(3, 3)$  and  $(7, 5)$  is a diameter of circle  $C$ .  $C$  is translated 3 left, reflected in the  $x$ -axis, then enlarged scale factor 4 about its centre. What is the equation of the final circle? (*after TMUA Specimen Paper 1 Q9*)

- A)  $(x-2)^2 + (y-4)^2 = 80$   
 B)  $(x-2)^2 + (y+4)^2 = 80$   
 C)  $(x-2)^2 + (y-4)^2 = 320$   
 D)  $(x-2)^2 + (y+4)^2 = 320$

**Answer:** B)  $(x-2)^2 + (y+4)^2 = 80$

**Method:** Original centre  $(5, 4)$ ,  $r^2 = (2)^2 + 1^2 = 5$ . Translate 3 left:  $(2, 4)$ , reflect in  $x$ -axis:  $(2, -4)$ , enlarge  $4\times$  about centre:  $r^2 = 5 \cdot 16 = 80$ , so  $(x-2)^2 + (y+4)^2 = 80$ .

**Investigate further:** Enlargement about the centre leaves the centre fixed and scales radius by 4. The 320 options would be  $r^2 \cdot 4$  too large.



3. Circle  $O$  radius 6,  $\angle POQ \geq \pi/2$ , area  $[POQ] = 9\sqrt{3}$ . For variable  $R$  on the major arc  $PQ$ , what is the *greatest possible* area of  $\triangle PRQ$ ? (*adapted from TMUA 2022 Paper 1 Q14*)

- A)  $18 + 9\sqrt{3}$   
 B)  $27\sqrt{3}$   
 C)  $27 + 9\sqrt{3}$   
 D)  $36 + 9\sqrt{3}$

**Answer:** B)  $27\sqrt{3}$

**Method:**  $[POQ] = \frac{1}{2}r^2 \sin \theta = 18 \sin \theta = 9\sqrt{3} \Rightarrow \sin \theta = \sqrt{3}/2, \theta = 120^\circ$ . Chord  $PQ = 6\sqrt{3}$ , maximal height 9, area  $27\sqrt{3}$ .

**Investigate further:** If  $\theta$  had been  $60^\circ$ , max would be  $9 + 18 = 27$ .

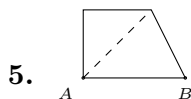
4. Let  $P(p, q)$  and circle  $C : x^2 + 2fx + y^2 + 2gy + h = 0$  with centre  $C_0 = (-f, -g)$  and  $L = |PC_0|$ . Which set is *minimal sufficient* to compute  $L$ ? (after MAT 2020 Q1H)

- A)  $f, g, h$
- B)  $f, g, p, q$
- C)  $f, h, p, q$
- D)  $g, h, p, q$

**Answer:** B)  $f, g, p, q$

**Method:**  $L^2 = (p + f)^2 + (q + g)^2$  needs  $f, g, p, q$ ;  $h$  irrelevant.

**Investigate further:** Tangent length needs  $h$ , not  $L$ .



- $ABCD$  is a cyclic quadrilateral with perpendicular diagonals intersecting at  $E$ . The line through  $E$  perpendicular to  $AB$  meets  $CD$  at  $F$ . Which statement is *necessarily true*? (adapted from BMO1 2010 Q4)

- A)  $F$  is the midpoint of  $CD$  (Brahmagupta)
- B)  $F$  bisects  $\angle CED$
- C)  $EF = AB/2$
- D)  $ABCD$  must be an isosceles trapezium

**Answer:** A)  $F$  is the midpoint of  $CD$  (Brahmagupta)

**Method:** Brahmagupta: In a cyclic quadrilateral with  $\perp$  diagonals, the perpendicular from the intersection to a side bisects the opposite side.

**Investigate further:** The other options fail for a general kite.

## Top 5 Patterns — Worksheet #2

- Complete the square to read centre and radius.** In  $x^2 + y^2 + 2gx + 2fy + c = 0$  the centre is  $(-g, -f)$  and  $r = \sqrt{g^2 + f^2 - c}$ . Strip the linear terms into squares before touching  $c$ . (see A1, A2, A4, B2, C8)
- Tangency = distance-to-line equals radius.** For a tangent,  $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$ . Never substitute line into circle. (see A9, B6, C1, D4)
- Tangent at a point: perpendicular to the radius / split variables.** On  $x^2 + y^2 = r^2$ , tangent at  $(x_1, y_1)$  is  $xx_1 + yy_1 = r^2$ . (see A7, B3, C4, D3)
- Chord: half-chord =  $\sqrt{r^2 - d^2}$ .** The perpendicular from the centre bisects the chord and makes a right triangle with half-chord, distance, and radius. (see A10, B7, B10, C5, D1)

- 5. Line–circle positioning by comparing  $d$  to  $r$ .**  $d < r$  secant,  $d = r$  tangent,  $d > r$  external.  
(see B5, C3, C8)

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**Common Traps — Worksheet #2**

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- 1. Forgetting to halve the linear coefficients.**  $x^2 + y^2 - 6x - 8y + 24 = 0$  has centre  $(3, 4)$ , not  $(-6, -8)$ . Centre is  $(-g, -f)$  where  $2g, 2f$  are the linear terms. (see A1, A4, B2)
- 2. Negative reciprocal error on the tangent slope.** Tangent to a circle is perpendicular to the radius, not a mere negation or inversion. (see A6, A7, B3)
- 3. Applying tangency when the point is not on the circle.** The tangent-at-a-point formula only holds for points on the circle; check  $x_1^2 + y_1^2 = r^2$  first. (see B3, C4, D3)
- 4. Generic lines through the centre are diameters, not tangents.** A line through the centre always cuts twice; tangency needs the perpendicular distance to be exactly  $r$ , not 0. (see B5, C3, C8)
- 5. Mixing up distance-to-centre with chord length.** The perpendicular distance  $d$  is one leg; the half-chord is the other. Squaring and subtracting, then doubling. (see B7, B10, C5, D1)

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*“Tangency is a distance statement in disguise: compare  $d$  to  $r$ , never solve the intersection.”*

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Found a mistake or want to contribute a sheet?  
Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)